

2. Moment of Inertia

2.1 Moment of Inertia of a plane area

Moment of inertia of a plane area (also known as second moment of the area) with respect to x-axis is defined by $I_{xx} = \int y^2 dA$ where an elemental area is multiplied by square of its distance from the axis and integrated over the entire area. Similarly, we can define

$$I_{yy} = \int x^2 dA$$

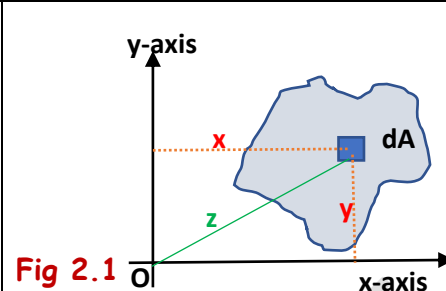


Fig 2.1

Perpendicular axis theorem

Consider the z-axis passing through point O and perpendicular to both x & y axes. By definition, moment of inertia of the area with respect to z-axis is

$$I_{zz} = \int z^2 dA = \int (x^2 + y^2) dA = \int y^2 dA + \int x^2 dA = I_{xx} + I_{yy}$$

Please note that the perpendicular axis theorem is valid only for plane areas and thin plates (plane laminae), not for three dimensional bodies.

Moment of inertia of an area about an axis perpendicular to its plane is known as **polar moment of inertia**

Parallel axis theorem

Consider a plane area whose centroid is at G. With respect to the co-ordinate axes X-Y passing through O, the centroid is at (a, b). Let us fix another co-ordinate axes x-y at point G. With respect to this centroidal axes, the centroid is at (0, 0). That is,

$$(\bar{X}, \bar{Y}) = (a, b)$$

$$(\bar{x}, \bar{y}) = (0, 0)$$

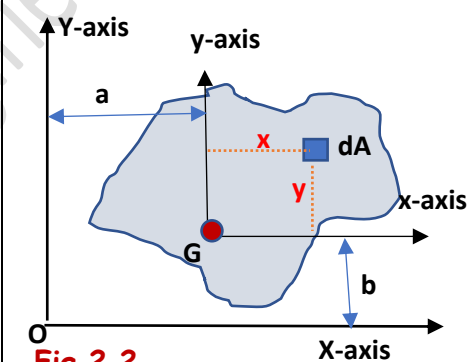


Fig 2.2

Consider an elemental area dA at distances x, y from the centroidal axes. Moment of inertia of the area with respect to X-axis is

$$\begin{aligned} I_{XX} &= \int (\text{distance from X axis})^2 dA = \int (b + y)^2 dA = \int y^2 dA + \int b^2 dA + \int 2b y dA \\ &= \int y^2 dA + b^2 \int dA + 2b \int y dA \end{aligned}$$

From the topic of centroid, we know that $\frac{\int y dA}{\int dA} = \bar{y}$, where $\bar{y}$ is the y-coordinate of the centroid of the area with respect to the axis from which the distance y is measured. Here y is measured from the x-axis through G. Therefore, $\bar{y} = 0$. Hence $\int y dA = 0$

$I_{XX} = \int y^2 dA + b^2 \int dA = \bar{I}_{xx} + b^2 A$ where $\bar{I}_{xx}$ is the moment of inertia of the area with respect to the parallel axis through its centroid. This is known as parallel axis theorem for moment of inertia which states that moment of inertia of an area with respect to any axis X is equal to the **sum of** (the moment of inertia of the area with respect to a parallel centroidal axis $\bar{x}$) **and** (the product of the area and the square of the distance between the parallel axes).

Similarly, we can write

$$I_{YY} = \int (a + x)^2 dA = \int x^2 dA + a^2 \int dA + 2a \int x dA = \bar{I}_{yy} + a^2 A$$

If we know the moment of inertia of an area with respect to an axis passing through its centroid, we can find out its moment of inertia with respect to any other parallel axis.

Please remember that when we use moment of inertia, the reference axis is very important.

Q1. Calculate the moment of inertia of a rectangular area having sides b and h with respect to the axes shown in figure.

A: Consider an infinitesimally small rectangular element of width dx and height y at a distance x from the y axis. The entire element is at a distance x from the y -axis. Hence by definition

$$I_{yy} = \int x^2 dA$$

Area of the infinitesimally small element, $dA = y \cdot dx = h \cdot dx$ (because $y=h$)

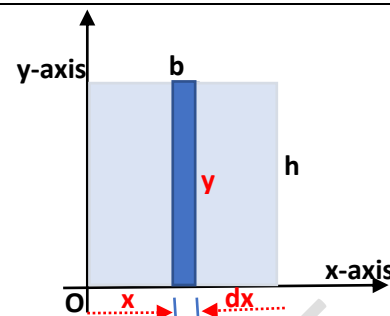


Fig 2.3

$I_{yy} = \int x^2 h dx = h \int_0^b x^2 dx = h \left[\frac{x^3}{3} \right]_0^b = \frac{b^3 h}{3}$ (to carry out integration over the entire area, the elemental strip needs to be moved from $x=0$ to $x=b$. Hence the limits of integration).

To find I_{xx} , we cannot use $I_{xx} = \int y^2 dA$, because the entire strip is not at the same distance y from the x -axis.

So, we need to consider another element of width dy and length x at a height y from the x -axis

$$I_{xx} = \int y^2 dA$$

Area of the infinitesimally small element, $dA = x \cdot dy = b \cdot dy$ (because $x=b$)

$I_{xx} = \int y^2 b dy = b \int_0^h y^2 dy = b \left[\frac{y^3}{3} \right]_0^h = \frac{bh^3}{3}$ (to carry out integration over the entire area, the elemental strip needs to be moved from $y=0$ to $y=h$. Hence the limits of integration).

The moments of inertia of a rectangular area with respect to the x - y axes passing through its edges are given by

$$I_{xx} = \frac{bh^3}{3} \quad I_{yy} = \frac{b^3 h}{3}$$

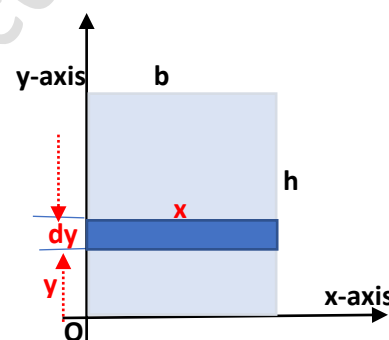


Fig 2.4

Now we can find out the moment of inertia of the rectangular area with respect to its centroidal x - y axes by applying the parallel axis theorem.

$A = bh$, $I_{xx} = \frac{bh^3}{3}$, $I_{yy} = \frac{b^3 h}{3}$, the centroid of the rectangle is at $(b/2, h/2)$ from the edges. The distance between the parallel horizontal axes is $h/2$ and that between the parallel vertical axes is $b/2$.

$$I_{XX} = \bar{I}_{xx} + \left(\frac{h}{2}\right)^2 A$$

$$\bar{I}_{xx} = \frac{bh^3}{3} - \left(\frac{h}{2}\right)^2 (bh) = \frac{bh^3}{12}$$

$$\bar{I}_{yy} = \frac{b^3 h}{3} - \left(\frac{b}{2}\right)^2 (bh) = \frac{b^3 h}{12}$$

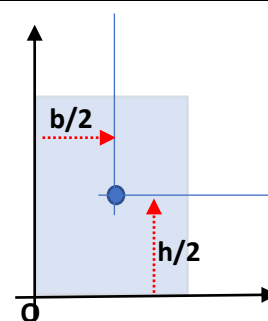


Fig 2.5

Q2. Calculate the moment of inertia of a right triangular area having sides b and h with respect to the axes shown in figure.

A: Consider an infinitesimally small rectangular element of width dx and height y at a distance x from the y axis. The entire element is at a distance x from the y -axis. Hence by definition

$$I_{yy} = \int x^2 dA$$

Area of the infinitesimally small element, $dA = y dx$

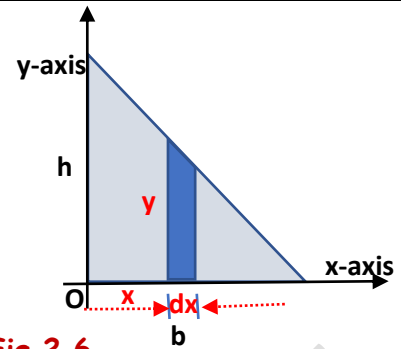


Fig 2.6

From the similar triangles, we can write $\frac{h}{y} = \frac{b}{b-x} \gg y = \frac{h}{b}(b-x)$

$$dA = \frac{h}{b}(b-x) dx$$

$I_{yy} = \int x^2 dA = \int_0^b x^2 \frac{h}{b}(b-x) dx$ (The integration is w.r.t x , the limits for x varies from 0 to b)

$$I_{yy} = \int_0^b \left(hx^2 - \frac{h}{b}x^3 \right) dx = \left[\frac{hx^3}{3} - \frac{h}{4b}x^4 \right]_0^b = \frac{b^3h}{12}$$

To find I_{xx} , we can consider an elemental strip parallel to x -axis as we have done in Q 1 (you can try it out). But we are going to use the same strip in Fig. 2.6 to find out I_{xx} .

From Q 1, we found out that the moment of inertia of a rectangular area with respect to the axis passing through the bottom edge is given by $\frac{bh^3}{3}$. Therefore, the moment of inertia of the elemental rectangular strip with respect to the x -axis passing through its bottom edge will be $\frac{dx \cdot y^3}{3}$. Hence the moment of inertia of the whole area with respect to the x -axis will be

$$I_{xx} = \int \frac{y^3 dx}{3} = \int_0^b \frac{\left(\frac{h}{b}(b-x) \right)^3 dx}{3} = \frac{h^3}{3b^3} \int_0^b (b^3 - 3b^2x + 3bx^2 - x^3) dx$$
 (The integration is w.r.t x , the limits for x varies from 0 to b)

$$I_{xx} = \frac{bh^3}{12}$$

Let us find out the moment of inertia of the right triangular area with respect to its centroidal x - y axes using the parallel axis theorem.

$A = \frac{1}{2}bh$, $I_{xx} = \frac{bh^3}{12}$, $I_{yy} = \frac{b^3h}{12}$, the centroid of the rectangle is at $(b/3, h/3)$ from the edges that are at right angle. The distance between the parallel horizontal axes is $h/3$ and that between the parallel vertical axes is $b/3$.

$$I_{XX} = \overline{I}_{xx} + \left(\frac{h}{3} \right)^2 A$$

$$\overline{I}_{xx} = \frac{bh^3}{12} - \left(\frac{h}{3} \right)^2 \left(\frac{1}{2}bh \right) = \frac{bh^3}{36}$$

$$\overline{I}_{yy} = \frac{b^3h}{12} - \left(\frac{b}{3} \right)^2 \left(\frac{1}{2}bh \right) = \frac{b^3h}{36}$$

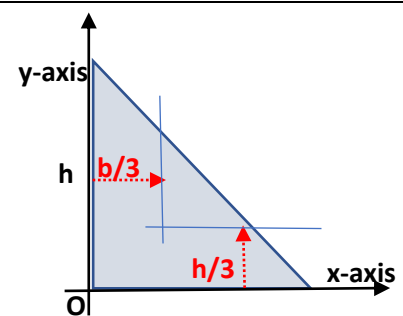


Fig 2.7

Q3. Calculate the moment of inertia of a quarter circular area having centre at the origin and radius r.

A: Consider an infinitesimally small rectangular element of width dx and height y at a distance x from the y axis. The entire element is at a distance x from the y -axis. Hence by definition $I_{yy} = \int x^2 dA$

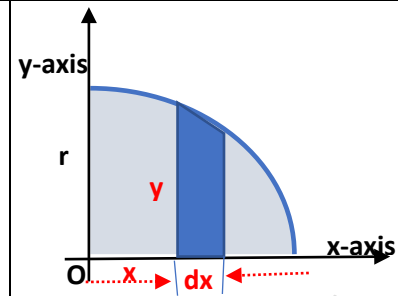


Fig 2.8

Area of the infinitesimally small element, $dA = y dx$

From the equation of a circle $x^2 + y^2 = r^2$, we can write $y = \sqrt{r^2 - x^2}$

$$dA = \sqrt{r^2 - x^2} dx$$

$I_{yy} = \int x^2 dA = \int_0^r x^2 \sqrt{r^2 - x^2} dx$ (The integration is w.r.t x , the limits for x varies from 0 to r)

$$I_{yy} = \int_0^{\frac{\pi}{2}} (r \sin \theta)^2 \sqrt{r^2 - (r \sin \theta)^2} r \cos \theta d\theta$$

$$= r^4 \int_0^{\frac{\pi}{2}} \sin^2 \theta \sqrt{1 - \sin^2 \theta} \cos \theta d\theta$$

$$= r^4 \int_0^{\frac{\pi}{2}} \sin^2 \theta \cos^2 \theta d\theta = r^4 \int_0^{\frac{\pi}{2}} \frac{\sin^2 2\theta}{4} d\theta = \frac{r^4}{4} \int_0^{\frac{\pi}{2}} \left(\frac{1 - \cos 2\theta}{2} \right) d\theta$$

$$I_{yy} = \frac{r^4}{8} \left[\theta + \frac{\sin 2\theta}{2} \right]_0^{\frac{\pi}{2}} = \frac{\pi r^4}{16}$$

Let us substitute $x = r \sin \theta$

$$dx = r \cos \theta d\theta$$

When $x=0$, $\theta = 0$

When $x=r$, $\theta = \frac{\pi}{2}$

The moment of inertia of the elemental rectangular strip with respect to the x -axis passing through its bottom edge will be $\frac{dx \cdot y^3}{3}$. Hence the moment of inertia of the whole area with respect to the x -axis will be $I_{xx} = \int \frac{y^3 dx}{3} = \frac{1}{3} \int_0^r (\sqrt{r^2 - x^2})^3 dx$

$$I_{xx} = \frac{1}{3} \int_0^{\frac{\pi}{2}} (\sqrt{r^2 - r^2 \sin^2 \theta})^3 r \cos \theta d\theta =$$

$$\frac{1}{3} r^4 \int_0^{\frac{\pi}{2}} \cos^4 \theta d\theta$$

$$= \frac{1}{3} r^4 \left[\frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2} \right]$$

$$I_{xx} = \frac{\pi r^4}{16}$$

Let us substitute $x = r \sin \theta$

$$dx = r \cos \theta d\theta$$

When $x=0$, $\theta = 0$

When $x=r$, $\theta = \frac{\pi}{2}$

Use the reduction formula

$$\int_0^{\frac{\pi}{2}} \cos^m \theta d\theta = \frac{m-1}{m} \cdot \frac{m-3}{m-2} \cdots \frac{1}{2} \cdot \frac{\pi}{2} \text{ if } m \text{ is even}$$

#Exercise 1 Determine the moment of inertia of the quarter circular area with respect to its centroidal x - y axes.

Let us find out the **polar moment of inertia** of a circular area.

Let us consider a circular area of radius r . Let the z -axis passes through O . Consider an infinitesimally small elemental ring of thickness $d\rho$ at a distance ρ from the z -axis. By definition,

$$I_{zz} = \int \rho^2 dA = \int_0^r \rho^2 (2\pi \rho d\rho) = \frac{\pi r^4}{2}$$

$$dA = 2\pi \rho d\rho$$

$I_{zz} = I_{xx} + I_{yy}$ (Perpendicular axis theorem)

Since the area is symmetrical about both x - y axes, $I_{xx} = I_{yy}$

$$I_{xx} = I_{yy} = \frac{I_{zz}}{2} = \frac{\pi r^4}{4}$$

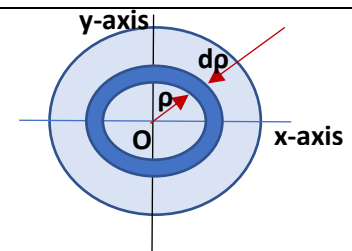


Fig 2.9

Q4. Calculate the moment of inertia of the area formed by a parabola $y=kx^2$ and a straight-line $y=x$ with respect to the x-y axes.

A: At the common point A (a,a), the equation of parabola becomes $a=ka^2$. Therefore, $k=1/a$.

The equation of parabola becomes $y=x^2/a$.

Consider an infinitesimally small rectangular element of width dx and height y at a distance x from the y-axis.

The entire element is at a distance x from the y-axis.

Hence by definition $I_{yy} = \int x^2 dA$

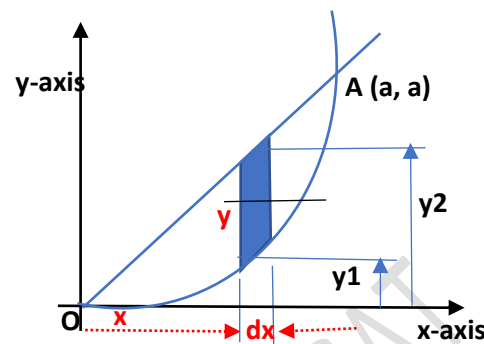


Fig 2.10

Area of the infinitesimally small element, $dA = y dx$

The element height $y = y_2 - y_1 = y$ co-ordinate of the line - y co-ordinate of the parabola

$$y = x - \frac{x^2}{a}$$

$$dA = y dx = \left(x - \frac{x^2}{a}\right) dx$$

$$I_{yy} = \int x^2 dA = \int x^2 \left(x - \frac{x^2}{a}\right) dx = \int_0^a \left(x^3 - \frac{x^4}{a}\right) dx = \frac{a^4}{20} \quad \text{(The integration is w.r.t x, the limits for x varies from 0 to a)}$$

The x-axis does not pass through the bottom edge of the elemental strip. Hence, we cannot use the formula $\frac{dx \cdot y^3}{3}$ here. Let us find out the moment of inertia of the elemental strip with respect to the x-axis by applying the parallel axis theorem. We know that the moment of inertia of the rectangular area with respect to its centroidal x-axis is $\frac{bh^3}{12} = \frac{dx \cdot y^3}{12}$. The distance between the

centroidal x-axis and the given x-axis is $y_1 + \frac{y}{2} = \frac{x^2}{a} + \frac{x - \frac{x^2}{a}}{2} = \frac{x^2}{a} + \frac{x}{2} - \frac{x^2}{2a} = \frac{x^2}{2a} + \frac{x}{2} = \frac{1}{2} \left(x + \frac{x^2}{a}\right)$

The moment of inertia of the differential element with respect to the x-axis

$$dI_{xx} = \overline{dI_{xx}} + \left(y_1 + \frac{y}{2}\right)^2 dA = \frac{y^3 dx}{12} + \left\{\frac{1}{2} \left(x + \frac{x^2}{a}\right)\right\}^2 \left(x - \frac{x^2}{a}\right) dx$$

Therefore, the moment of inertia of the total area with respect to x-axis

$$I_{xx} = \int dI_{xx} = \int \frac{\left(x - \frac{x^2}{a}\right)^3 dx}{12} + \int \left\{\frac{1}{2} \left(x + \frac{x^2}{a}\right)\right\}^2 \left(x - \frac{x^2}{a}\right) dx$$

$$= \frac{1}{12} \int_0^a \left(x^3 - 3x^2 \frac{x^2}{a} + 3x \frac{x^4}{a^2} - \frac{x^6}{a^3}\right) dx + \frac{1}{4} \int_0^a \left(x^3 + \frac{x^4}{a} - \frac{x^5}{a^2} - \frac{x^6}{a^3}\right) dx$$

$$I_{xx} = \frac{a^4}{28}$$

#Exercise 2

Find the moment of inertia of the area with respect to the given x-y axes.

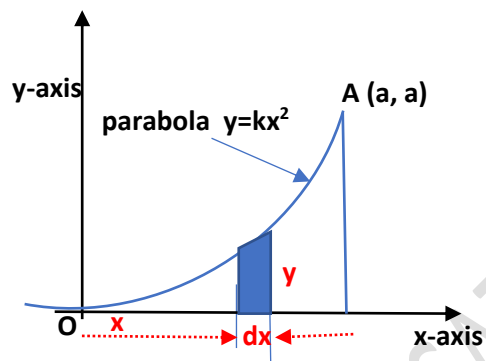


Fig 2.11

#Exercise 3

Find the moment of inertia of the area with respect to the given x-y axes.

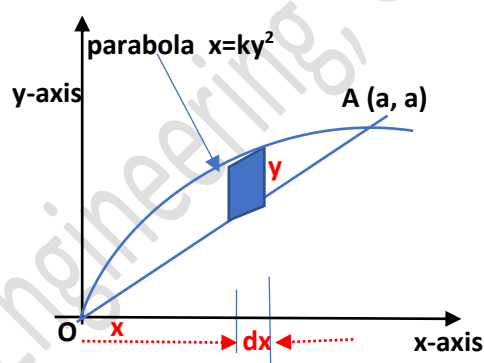


Fig 2.12

Prof. Biju N., School of Engineering, CUSAT